

Step-by-Step Project Development Procedure

This document provides a comprehensive overview of the workflow executed to develop the "Strategic Product Placement Analysis" project.

1. Problem Identification

Objective: To determine how different product placements (Aisle, End-cap, Front of Store) impact sales volume and how demographic and seasonal factors influence these outcomes. Retail managers struggle to optimize floor space without clear visual data.

2. Data Collection

- **Source:** Kaggle (Dataset: "Impact of Product Positioning on Sales").
- **Process:** Downloaded the raw data in CSV format.
- **Validation:** Ensured the dataset contained the required records without null values or corrupt entries before ingestion into the BI tool.

3. Data Cleaning & Preparation

- **Ingestion:** Connected the CSV to Tableau Desktop.
- **Transformation:** Renamed columns for readability and validated data structures.
- **Calculation Fields:** Created specific calculated metrics required for analysis:
 - Price
 - Competitors Price
 - Sales Volume
- **Verification:** Ensured data types were correctly assigned (String for categories, Geographic for demographics if applicable, and Numeric for financial metrics).

4. Data Visualization & Dashboarding

- **Worksheet Generation:** Created exactly 8 distinct visualizations to cover all analytical angles:
 1. Avg Sales Volume by Product Category
 2. Avg Sales Volume by Product Category on Sales
 3. Consumer Demographics vs Product Category
 4. Foot traffic vs Sales Volume
 5. Competitors Price vs Price
 6. Promotion of Product Category on Price and Sales Volume
 7. Product Category vs Price

8. Avg Sales Volume by Product Category by Product Position

- **Dashboard Assembly:** Aggregated all 8 charts into a single interactive dashboard.
- **Filters Implementation:** Configured global filters to refine and focus on specific subsets of data relevant to the analysis objectives.

5. Result Generation (Storyboarding)

Synthesized the findings into a 3-scene Tableau Story:

- **Scene 1 Results:** Identified that the "Clothing" category yields the highest average sales volume overall.
- **Scene 2 Results:** Discovered that product positions at the "Front of store" correspond directly to "High sales volume" and "High foot traffic".
- **Scene 3 Results:** Concluded that optimizing "Clothing" placement specifically at the "Front of store" maximizes revenue potential.

6. Web Integration & Publishing

- **Publishing:** Saved the completed Tableau workbook to Tableau Public.
- **Web UI Construction:** Developed a custom web application using Python (Flask) and a Bootstrap HTML template.
- **Integration:** Embedded the Tableau Public Dashboard and Story scenes into the web application, making the insights accessible to non-technical stakeholders via a standard web browser interface.